C

CAMBRIDGE ADDITIONAL MATHEMATICS (0606) • UNIT 7.01

Differentiation & Rates of Change

PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Differentiation & Rates of Change Practice Exercises

1 MCQ

Find the equation of the tangent line to the curve $y = (2x + 1)^3$ at the point where $x = 0$.

A $y = 2x + 1$

B $y = 6x - 1$

C $y = 6x + 1$

D $y = 6x + 3$

2 MCQ

Find the equation of the tangent line to the curve $y = (3x + 1)^3$ at the point where $x = 0$.

A $y = 9x + 3$

B $y = 9x - 1$

C $y = 3x + 1$

D $y = 9x + 1$

3 MCQ

Find the equation of the tangent line to the curve $y = (4x + 1)^3$ at the point where $x = 0$.

A $y = 12x + 3$

B $y = 12x - 1$

C $y = 4x + 1$

D $y = 12x + 1$

4 MCQ

Find the equation of the tangent line to the curve $y = (5x + 1)^3$ at the point where $x = 0$.

A $y = 15x + 3$

B $y = 15x - 1$

C $y = 5x + 1$

D $y = 15x + 1$

5 MCQ

Find the equation of the tangent line to the curve $y = (6x + 1)^3$ at the point where $x = 0$.

A $y = 18x + 1$

B $y = 6x + 1$

C $y = 18x - 1$

D $y = 18x + 3$

6 MCQ

Find the equation of the tangent line to the curve $y = (7x + 1)^3$ at the point where $x = 0$.

A $y = 21x - 1$

B $y = 7x + 1$

C $y = 21x + 3$

D $y = 21x + 1$

7 MCQ

Find the equation of the tangent line to the curve $y = (8x + 1)^3$ at the point where $x = 0$.

A $y = 8x + 1$

B $y = 24x + 3$

C $y = 24x - 1$

D $y = 24x + 1$

8 MCQ

Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 3$ cm?

A 18π cm^3/s

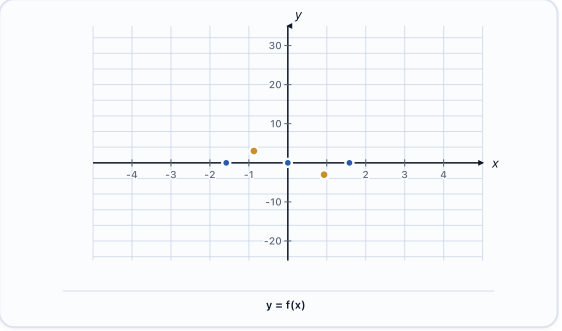
B 36π cm^3/s

C 9π cm^3/s

D $\frac{18}{2}\pi$ cm^3/s

9 MCQ

The variables x and y are related by the equation $y = 2x^3 - 5x$. Given that x is increasing at a constant rate of 3 units per second, find the rate of change of y with respect to time at the instant when $x = 2$.



A 19 units per second

B $\frac{19}{3}$ units per second

C 21 units per second

D 57 units per second

10 MCQ

Let $f(x) = \dfrac{140x^2 + 10x - 10}{x^2 + 14}$ for $x > 0$. Find the limit at infinity $\lim\limits_{x \to \infty} f(x)$.

A 143

B 126

C 150

D 140

11 MCQ

A storage capsule consists of a right circular cylinder of radius r cm and height h cm, closed with a flat circular base of radius r cm at the bottom, and surmounted by a hemispherical dome of radius r cm at the top. The total exterior surface area of the capsule is fixed at 180π cm^2.
During a heating process, the radius r expands at a constant rate of 0.4 cm s^{-1}. Find the rate of change of the total volume of the capsule, in cm^{3}s^{-1}, at the instant when the height of the cylindrical section is 16.5 cm .

A 50π

B $\frac{36\pi}{5}$

C 20π

D 11π

12 MCQ

The variables x and y are connected by the equation $y = \sqrt{2x+5}$. Given that x increases at a constant rate of 6 units per second, find the rate of change of y with respect to time when $x = 2$.

A 1 unit per second

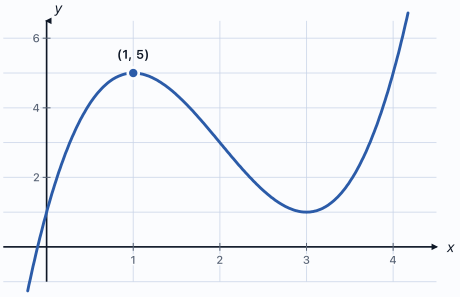
B $\frac{1}{3}$ units per second

C 2 units per second

D 12 units per second

13 MCQ

The diagram shows part of the curve $y = ax^3 + bx^2 + 9x + 1$, where a and b are constants. The curve has a stationary point at $(1, 5)$.
Find the coordinates of the other stationary point on the curve.



A $(3, 0)$

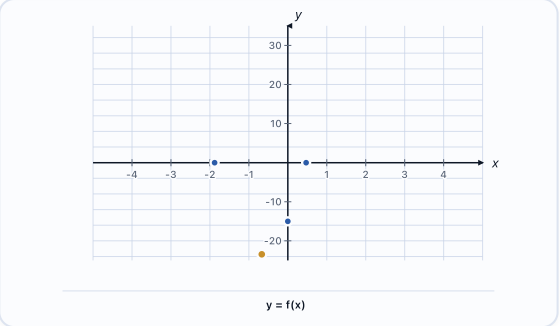
B $(3, 1)$

C $(3, -1)$

D $(-3, 1)$

14 MCQ

Let the differentiable function f be given by $f(x) = 17x^2 + 24x - 15$. Find the derivative $f'(15)$.



A 534

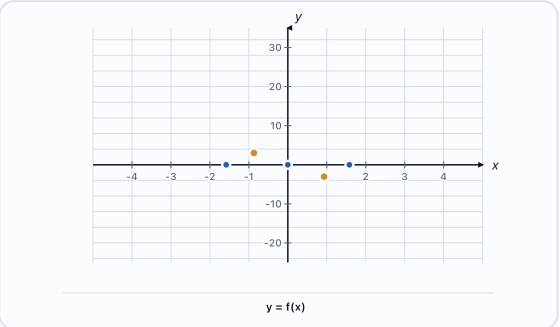
B 510

C 551

D 537

15 MCQ

The variables x and y are connected by the equation $y = 2x^3 - 5x$. Given that x increases at a constant rate of 4 units s^{-1}, find the rate of change of y at the instant when $x = 2$.



A 76 units s^{-1}

B 24 units s^{-1}

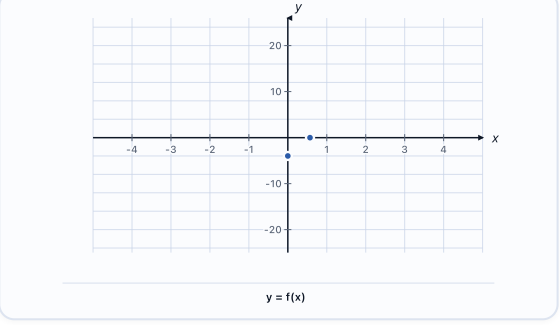
C 19 units s^{-1}

D 28 units s^{-1}

16 MCQ

The line $y = 7x - 4$ is tangent to the curve $y = ax^2 + \frac{b}{x}$ at the point where $x = 2$, where a and b are constants and $x \neq 0$.

Find the coordinates of the stationary point of the curve and determine its nature.



A $(1, 6)$, which is a local maximum

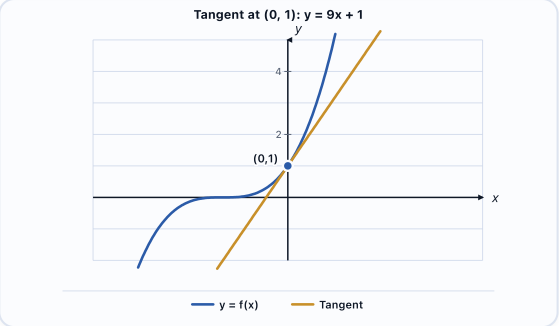
B $(1, 6)$, which is a local minimum

C $(-1, -2)$, which is a local minimum

D $(2, 10)$, which is a local minimum

17 MCQ

Calculate the equation of the tangent line to the curve $y = (3x + 1)^3$ at the point where $x = 0$.



A $y = 9x + 1$

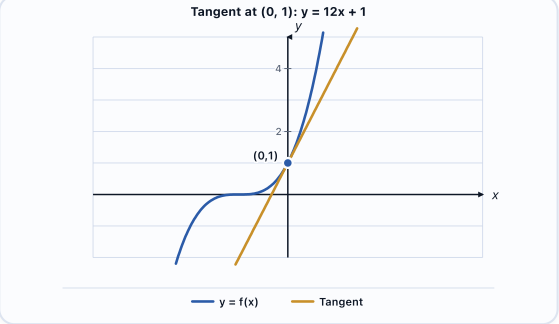
B $y = 9x + 3$

C $y = 9x - 1$

D $y = 3x + 1$

18 MCQ

Compute the equation of the tangent line to the curve $y = (4x + 1)^3$ at the point where $x = 0$.



A $y = 12x + 1$

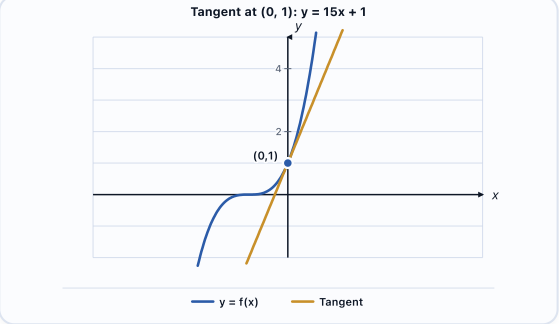
B $y = 12x + 3$

C $y = 12x - 1$

D $y = 4x + 1$

19 MCQ

Deduce the equation of the tangent line to the curve $y = (5x + 1)^3$ at the point where $x = 0$.



A $y = 5x + 1$

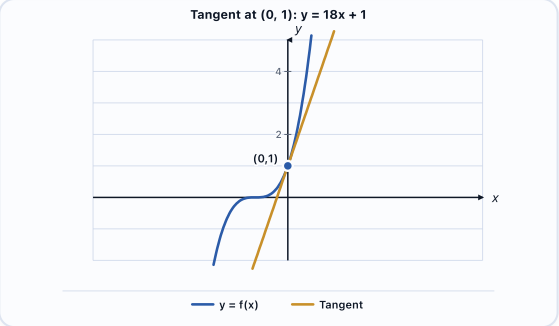
B $y = 15x + 3$

C $y = 15x + 1$

D $y = 15x - 1$

20 MCQ

Obtain the equation of the tangent line to the curve $y = (6x + 1)^3$ at the point where $x = 0$.



A $y = 6x + 1$

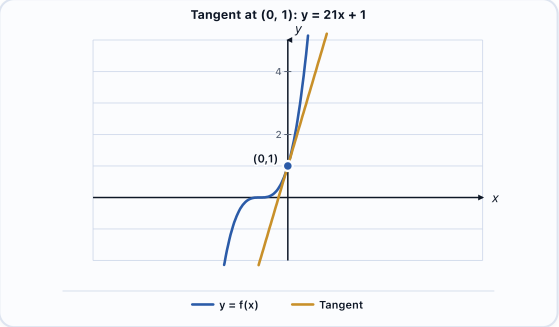
B $y = 18x - 1$

C $y = 18x + 3$

D $y = 18x + 1$

21 MCQ

Identify the equation of the tangent line to the curve $y = (7x + 1)^3$ at the point where $x = 0$.



A $y = 21x + 1$

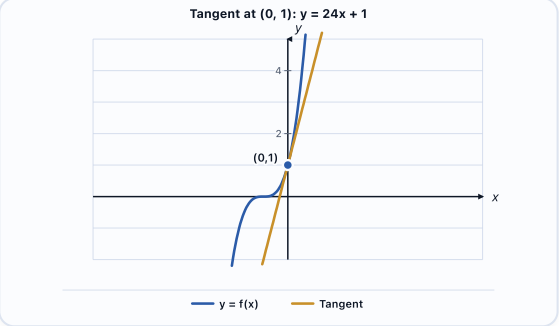
B $y = 21x + 3$

C $y = 21x - 1$

D $y = 7x + 1$

22 MCQ

Establish the equation of the tangent line to the curve $y = (8x + 1)^3$ at the point where $x = 0$.



A $y = 24x - 1$

B $y = 8x + 1$

C $y = 24x + 3$

D $y = 24x + 1$

23 MCQ

Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 4$ cm?

Given Motion Parameters

Parameter 1	Parameter 2
0.5	4

A 32π cm^3/s

B $\frac{32}{2}\pi$ cm^3/s

C 64π cm^3/s

D 16π cm^3/s

24 MCQ

Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 5$ cm?

Given Motion Parameters

Parameter 1	Parameter 2
0.5	5

A 50π cm^3/s

B 25π cm^3/s

C 100π cm^3/s

D $\frac{50}{2}\pi$ cm^3/s

25 MCQ

Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 6$ cm?

Given Motion Parameters

Parameter 1	Parameter 2
0.5	6

A $\frac{72}{2}\pi$ cm^3/s

B 144π cm^3/s

C 36π cm^3/s

D 72π cm^3/s